



NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Third Semester B.Tech (CBCS) Degree Examinations
 December 2024

CS2001-1 – DATA STRUCTURES
 (For AD, AM, CB, CC, CS, IS)

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II** each and **One full** question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS**20 Marks**

- * and & operators are respectively**
 - Dereferencing and Memory operators
 - Address of and Dereferencing operators
 - Dereferencing and Address of operators
 - Memory operator and address of operators
- What does the following array declaration imply?**
`int *arr[10];`
 - Array of 10 integers
 - Array of 10 pointers to integers
 - Pointer to an array of 10 integers
 - Array of 10 Characters
- What does the following structure declaration imply?**

```
struct S {
    int a[3];
    struct S *next;
};
```

 - A structure containing an array of 3 integers and a pointer to an integer
 - A structure containing an array of 3 integers and a pointer to another structure of the same type
 - A structure containing a pointer to an array of integers
 - A structure containing an array of 3 integers and a variable next
- What will be the output of the following C code?**

```
#include <stdio.h>
int main()
{
    int *ptr, a = 10;
    ptr = &a;
    *ptr += 1;
    printf("%d,%d\n", *ptr, a);
}
```

 - 11,11
 - 11,10
 - 10,10
 - 10,11
- In a stack, if a user tries to remove an element from empty stack it is called**
 - Underflow
 - Overflow
 - Empty collection
 - Garbage Collection
- Consider the following stack operations: push(5), push(10), push(20), pop(), push(15), pop(). What will be the final state of the stack?**
 - [5, 10, 20, 15]
 - [5, 10, 15]
 - [5, 15]
 - [5, 10]
- Which is the most appropriate data structure for reversing a word?**
 - Queue
 - Array
 - Stack
 - Pointers

8. Evaluate the postfix expression $6\ 2/5 - 42 + *$ Answer is
 A) 8 B) 10
 C) 12 D) 14
9. If the elements "A", "B", "D" and "C" are placed in a queue and are deleted one at a time, in what order will they be removed?
 A) DCBA B) ABDC
 C) ABCD D) CDBA
10. A normal queue, if implemented using an array of size MAX_SIZE, gets full when
 A) Rear=front B) Front=(rear+1)mod MAX_SIZE
 C) Front=rear+1 D) Rear=MAX_SIZE-1
11. An array of size MAX_SIZE is used to implement a circular queue. Front, Rear, and count are tracked. Suppose front is 0 and rear is MAX_SIZE -1. How many elements are present in the queue?
 A) Zero B) One
 C) MAX_SIZE-1 D) MAX_SIZE
12. In which Queue deletion operations performed deleting the largest number?
 A) Double ended queue B) Ascending Priority Queue
 C) Descending Priority Queue D) Circular Queue
13. Which of the following is true about the head node in a singly linked list?
 A) It contains the largest value in the list B) It points to the first node of the list
 C) It points to the last node of the list D) It is always null
14. Consider the following definition in c programming language.

```
struct node
{
  int data;
  struct node * next;
}
typedef struct node NODE;
NODE *ptr;
```

 Which of the following is correct statement
 A) ptr = (NODE*)malloc(sizeof(NODE)); B) .ptr = (NODE*)malloc(NODE);
 C) ptr = (NODE*)malloc(sizeof(NODE*)); D) ptr = (NODE)malloc(sizeof(NODE));
15. In a Circular Singly Linked List, to delete a node at the beginning, the pointer of which node must be updated?
 A) first node B) Second node
 C) Last node D) The node to be deleted
16. Which of the following operations is more efficient in a doubly linked list than in a singly linked list?
 A) Searching for an element B) Deleting a node when the pointer to the node is given
 C) Traversing the list D) Accessing the first element
17. The number of nodes in a fully binary tree of level 5 is -----
 A) 64 B) 67
 C) 32 D) 63
18. In a binary tree, the preorder traversal of nodes results in the sequence 4, 2, 5, 1, 3. What is the root of the binary tree?
 A) 1 B) 2
 C) 3 D) 4
19. In an expression tree, what does the leaf node represent?
 A) Operator B) Operand
 C) Root of the tree D) Expression
20. What is the maximum number of edges in a simple undirected graph with n vertices?
 A) $n * (n - 1)$ B) $n * (n - 1)/2$
 C) $n * (n + 1)$ D) $n * (n + 1)/2$

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1. a)	With syntax explain memory management functions used in C to allocate and release memory during run time in C	6	L*2	1	1
b)	Design a C program to enter details of n students Roll_No, Name, marks in five subjects and display the same. Write function to compute the total marks of individual student and display the same	6	L3	1	3
c)	With suitable example show the address calculation to access the elements in a 2D array	4	L3	1	3
2. a)	Design an integer stack S using structure and write C functions for primitive operations of stack for the function calls Push(&S, value), Pop(&S) and display(&s) with proper testing conditions to avoid illegal operations	6	L3	2	3
b)	Covert the infix expression $(4+8)*(6-5)/((3-2)*(2+2))$ to a suffix expression and show the evaluation of that suffix expression using evaluation algorithm	6	L3	2	3
c)	Convert the following infix expressions to prefix and postfix expressions: (i) $(A-B)/C*(D+E-F/G)$ (ii) $(A+B)*(C-D)/E+F$	4	L3	2	3
3. a)	Develop infix-to-postfix conversion algorithm and apply the same to convert the following infix expression to postfix expression using stack $A*(B+C)/D$	6	L3	2	3
b)	Give recursive definitions to (i) Compute factorial of n (ii) nth Fibonacci number (iii) GCD(m,n) to find gcd Also Write recursive algorithm to Solve the Tower of Hanoi problem.	6	L3	2	3
c)	Define recursion? What do you mean by well-defined recursion? What is a recursive chain?	4	L2	2	1
Unit – II					
4. a)	With a neat diagram explain how circular queue can be implemented using an array? Write C functions Insert and delete to demonstrate working of Circular queue	6	L3	3	3
b)	What is singly linked list? With a proper node structure write C functions to insert a node and delete a node from the front of list?	6	L3	4	3
c)	Design a C functions to display each node information and also the total number of nodes present in that given linked list	4	L3	4	3
5. a)	Define Queue data structure. Develop a menu driven C program to demonstrate the working of ordinary queue that uses the pointer to structure concept.	6	L3	3	3
b)	Write C functions to perform the following operation in a singly linked list (i) Insert a node with data value x after a node whose data value is y (ii) delete a node following a given node whose data value is y	6	L3	4	3
c)	Write C function to have an ordered list in a singly linked list	4	L3	4	3
6. a)	What is circular linked list? With a proper node structure give C functions to create a circular linked list with 5 nodes by front insertion and display that circular list	6	L3	4	3

- b) What is doubly linked list? With proper node structure give C functions to insert a new node to the right of a node and to delete a given node
- c) Design C functions to implement a Queue using circular linked list

6	L3	4	3
4	L3	4	3

Unit – III

7. a) Define the following:
i) Binary tree ii) Fully binary tree iii) Complete binary tree
With example show a suitable method to represent a complete binary tree in C
- b) What is an expression tree? Write expression tree for the expression $A/(B*C)*(D+E)$. Write the C functions for preorder, inorder and postorder traversal for a binary tree and give the result of that traversals for the constructed expression tree.
- c) What is Threaded Binary Tree? Mention the rules to construct the in threaded binary tree. Give the pictorial representation of a in threaded complete binary tree having 9 nodes

6	L2	5	3
6	L3	5	3
4	L3	5	3

8. a) Define a Binary Search Tree (BST)? Construct a BST for the given list 14, 15, 4, 9, 7, 18, 3, 5, 16, 20, 17. Show how you can use this BST to sort the given list. Write recursive C function to search for a key in a given BST.
- b) What is a graph? explain two methods to represent the graph. Represent an undirected complete graph G with 4 vertices in both the methods.
- c) What are the two techniques used for collision resolution in hashing? For the example given below show the collision resolution in both methods

6	L3	5	3
6	L3	5	3
4	L2	5	3

- Record: A, B, C, D, E, X, Y, Z
- H(k): 4, 8, 2, 11, 4, 11, 5, 1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
